ICS / OT SECURITY

ICS/OT Security Assessment

Safety-first assessment of SCADA, PLCs, and industrial networks — aligned to IEC 62443.

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KEY BENEFITS

Why This Matters

Safety-First Testing
Non-disruptive methodologies designed for production OT environments.

Full IT-to-OT Chain
We test the complete attack path from corporate network to control systems.

IEC 62443 Aligned
Security level evaluation mapped to ISA/IEC 62443 for compliance-ready output.

YOUR CHALLENGE

Your industrial control systems manage physical processes critical to operations and safety:

- Can attackers reach control systems from the corporate network?
- Are IT/OT segmentation controls effective?
- What protocol vulnerabilities exist in Modbus, DNP3, OPC UA?
- Do PLCs and RTUs have exploitable weaknesses?

Why Technieum as Your Trusted Partner
Technieum brings 6 proprietary AI platforms, an in-house R&D team actively hunting zero-days, and AI-powered methodologies that deliver validated, exploitable findings with guaranteed zero false positives — enabling your team to focus remediation effort where it matters most.

OUR SOLUTION

Services Overview

Technieum conducts safety-first ICS/OT assessments using passive reconnaissance, controlled testing, and IEC 62443 evaluation.

1	**ICS/SCADA Assessment**	SCADA architecture, RTU/PLC access controls, HMI vulnerabilities, command injection.
2	**OT Architecture Review**	Purdue Model compliance, zone/conduit architecture, DMZ configs, data diodes.
3	**IT/OT Segmentation Testing**	Validates lateral movement paths from corporate IT into control systems.
4	**Protocol Security**	Modbus TCP, DNP3, OPC UA authentication, man-in-the-middle, replay attacks.
5	**PLC/RTU/DCS Testing**	Firmware analysis, default credentials, logic download authorization, config backup security.
6	**ICS Threat Modelling**	MITRE ATT&CK for ICS-mapped threat analysis with safety impact prioritization.

THE TECHNIEUM EDGE

Why Technieum for ICS/OT

Our Methodology
Safety planning → passive recon → architecture review → config assessment → controlled testing → IEC 62443 mapping → safety-validated remediation.

Safety-First
Every procedure designed for production OT — passive recon, controlled probing.

IT-to-OT Attack Chain
Complete path from corporate network through DMZ to control systems.

Protocol Expertise
Modbus, DNP3, OPC UA — authentication and injection testing.

IEC 62443 Alignment
Security level evaluation and zone/conduit validation.

Standards Alignment

Every Technieum engagement aligns with globally recognized security frameworks including OWASP Top 10, OWASP API Top 10, OWASP LLM Top 10, PTES, NIST CSF, NIST AI RMF, MITRE ATT&CK, CIS Benchmarks, IEC 62443, and ISO 27001:2022. Our team holds OSCP, OSEP, CRTE, CRTP, eWPTX, eCPPT, and CEH certifications. Technieum is ISO 9001:2015 and ISO 27001:2022 certified.

OUR METHODOLOGY

How We Deliver Results

Technieum's ICS/OT security methodology follows a safety-first five-phase approach designed for production industrial environments — every procedure is validated for non-disruptive execution before deployment.

Discover	Analyze	Attack	Report	Remediate
Asset enumeration	ToIP intelligence	Manual exploitation	CVSS/CWE scoring	AI-powered fixes
Tech fingerprinting	Logic extraction	Business logic test	PoC evidence	Stack-specific code
OSINT gathering	Threat modelling	Chain validation	Exec summary	Config guidance
Scope validation	Risk prioritization	Privilege escalation	Test case docs	Retest validation
Port scanning	Attack planning	Lateral movement	Playbook docs	Closure sign-off
Service detection	Playbook generation	Data exfiltration	Risk heatmap	Knowledge transfer

Phase 1 — Discover

Passive reconnaissance through network traffic analysis, protocol identification, and device fingerprinting — without active probing. Asset inventory of PLCs, RTUs, DCS, HMIs, historians, and engineering workstations. Safety constraint documentation.

Phase 2 — Analyze

Architecture review evaluates Purdue Model compliance, zone and conduit design, IT/OT DMZ configurations, and remote access pathways. IEC 62443 security level assessment. ToIP generates ICS-specific attack playbooks from OT vulnerability intelligence.

Phase 3 — Attack

Safety-validated controlled testing of IT/OT boundary controls, jump server security, protocol authentication (Modbus, DNP3, OPC UA), PLC/RTU configuration access, and HMI vulnerabilities. All testing follows pre-approved safety procedures.

Phase 4 — Report

ICS-specific technical reports with safety impact analysis, IEC 62443 compliance mapping, Purdue Model assessment results, protocol security findings, and MITRE ATT&CK for ICS TTP mapping. Risk prioritized by operational and safety impact.

Phase 5 — Remediate

OT-specific remediation with safety-validated implementation procedures — network segmentation rules, protocol authentication configuration, PLC hardening steps, and access control improvements. Change management-compatible retest procedures.

DELIVERABLES

What You Receive

Every engagement produces four comprehensive deliverables designed for different stakeholders — from technical teams requiring reproduction steps to executives needing business impact context.

Technical Report

CVSS-scored, CWE-mapped findings with proof-of-concept evidence and step-by-step reproduction instructions for every vulnerability discovered.

Management Report

Executive-level risk summary with business impact analysis, prioritized recommendations, and strategic remediation roadmap for leadership.

Threat Model Report

Comprehensive threat landscape analysis with attack vectors mapped to MITRE ATT&CK framework specific to your environment.

ToIP Playbook

Target-specific attack playbooks generated from 15,000+ real-world disclosed vulnerabilities via our RAG-powered intelligence engine.

- ✓ **Technical Report**
  Detailed findings with CVSS scoring, CWE mapping, proof-of-concept evidence, and step-by-step reproduction instructions.
- ✓ **Management Report**
  Executive-level risk summary with business impact analysis, prioritized recommendations, and strategic remediation roadmap.
- ✓ **Threat Model Report**
  Comprehensive threat landscape analysis specific to your environment, attack vectors, and adversary profiles mapped to MITRE ATT&CK.
- ✓ **ToIP Playbook Document**
  Target-specific attack playbooks generated from 15,000+ real-world disclosed vulnerabilities via our RAG-powered intelligence engine.

All deliverables are produced within the agreed timeline with milestone-driven tracking. Critical findings are escalated immediately via the OffSec Management Portal — you never wait for the final report to learn about urgent risks.

Ready to Secure Your Organization?

Partner with Technieum for AI-powered offensive security with zero false positives.

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